

News Video Generator - Project Report

Overview

This project is a Python-based script that automatically generates a short news video using real-time headlines. It fetches trending news, generates a script using a language model, converts the script into audio, creates relevant visuals using an AI image generator, and compiles everything into a final video.

Key Features

1. Real-time News Fetching

- Uses the **NewsAPI** to fetch the latest English-language headlines.
- Selects a specific article (6th in the list) for processing.

2. AI-Generated Script

- Generates a formal news script using the **DistilGPT2 model**.
- Adds an intro and outro for structure.
- Ensures the script reads like a typical news anchor briefing.

3. Text-to-Speech Conversion

- Uses **Google Text-to-Speech (gTTS)** to convert the script into audio.
- Outputs an .mp3 file to be synced with video.

4. Image Generation with Stable Diffusion

- Generates an AI image based on the news headline using **Stable Diffusion**.
- Ensures the visual matches the topic for better viewer engagement.

5. Slide Creation

- Creates three slides: **Intro**, **Main**, and **Outro**.
- Each slide is created with PIL, using the generated image and custom overlaid text.
- Handles font scaling and text wrapping.

6. Video Assembly Using FFmpeg

- Uses **FFmpeg** to:
 - Convert slides into short looping videos.
 - Concatenate them into a complete video.
 - Overlay the audio on the final video.

7. Cleanup Process

- Deletes temporary files like images, slides, intermediate videos, and audio files to keep the working directory clean.

Technologies Used

- **Python**
- **gTTS (Google Text-to-Speech)**
- **Pillow (PIL)**
- **NewsAPI**
- **Transformers (Hugging Face)**
- **Stable Diffusion (Diffusers)**
- **FFmpeg**

Execution Flow

1. Fetch news headline and description.
2. Generate script using a text-generation model.
3. Generate audio from script.
4. Create AI-generated image.
5. Make slides using PIL.
6. Convert slides to video using FFmpeg.
7. Merge videos and add audio.
8. Output the final video.
9. Cleanup.

Sample Output

- A short 60-second video that:
 - Begins with a "Breaking News" title screen.
 - Shows the headline and description on an AI-generated image.
 - Plays with audio that sounds like a news anchor's script.
 - Ends with a "Thanks for Watching" screen.